REFLECTION

After completing Activity 6.1, it can be helpful to think about how you approached it and ways you could improve your research skills for future activities.

1 Discuss the sources you used to find examples of robots. How did you make sure these sources were reliable, up-to-date and accurate? What would you do differently if you had to find further examples?

2 Discuss whether finding your own examples might help you answer exam questions. Is it helpful to find your own contexts instead of relying on those given in class?

Questions

6 Define the term robot.

7 Identify **two** ways robots can be used in industry.

8 Give **one** benefit and **one** drawback of the use of domestic robots.

9 Discuss the use of driverless cars.

6.4 Artificial intelligence

Can computers think? This is a question that has been asked and sought after for many years. Alan Turing developed a test in 1950, called the Turing Test, to assess whether a computer can display intelligence. The test involves a person holding a conversation with a computer program, and another person. If they cannot work out which is the computer and which is the person they are talking to, then it is has passed the Turing Test. Some people do not think that this would demonstrate intelligence, the computer is still following a set of rules that are pre-written.

What is artificial intelligence?

Artificial intelligence (AI) is the development of programs to simulate human behaviour. There is a broad range of areas within AI:

- Image recognition; to identify objects or people in an image.
- Speech recognition; to identify words spoken and store them.
- Natural language; to receive a command or instruction that is not in a set format and perform the required response.
- Computer games; to move elements or characters independently based upon the environment.
- Diagnosis systems; for example, medical diagnosis.

You have probably used natural language systems, for example, on a smartphone or a web browser. You can enter your request in dozens of different ways. Let's say you want to find out what time the sun sets today. You could enter "sunset today time", or "what

KEY WORD

artificial intelligence (AI): a part of computer science that looks at creating machines that can think and perform tasks a person would usually perform.

time does the sun set today?" or "today when does the sun set?" or many other ways. The software needs to understand what you have asked to decide how it will answer.

Some simple questions could be hard-coded. That means every possible way of asking a question is written in the program, so it just has to select the correct one. The problem is, how many ways are there? Is it possible to hard-code every possible question or statement that anyone in the world could ever give? This would be impossible. So instead, the artificial intelligence program has to be able to adapt to whatever is entered. This could be done by comparing it with similar phrases, or picking out key words and grammar, or it could use neural networks which attempt to simulate the human brain to process the words entered.

ACTIVITY 6.2

Identify any programs you have used that might make use of artificial intelligence. This could be computer games where you play against the computer, or natural language interfaces. Discuss how you think these may have been programmed and how the problems can be simplified into smaller parts to make the task easier to tackle.

Components of AI

Artificial intelligence programs can be simplified to the four key features:

1. **Collection of data:** A program will need data input, this might come from a user (e.g. speech recognition) or from its sensors (e.g. a robot that uses infra-red sensors to work out its position and where it can and can't move).
2. **A set of programmed rules**: These are stored for the program to use to make decisions. For example, a calculation to determine which decision to make. A rule that if an object is in front of the sensor, turn to the right.
3. **The ability to reason:** Reasoning is an area of logic, where you have rules, and from these you can develop facts. For example, all dogs eat meat. Fred is a dog. Therefore, Fred must eat meat.
4. **The ability to learn and adapt:** Not all AI systems will do this, this is a specialist area known as machine learning. Systems can be programmed to learn, and change their own rules and logic. This often involves training. If you want a program to differentiate between an image of a dog and a horse, you can train the system by showing it images of dogs and horses and telling it what the images are showing. The program will then learn the characteristics of a horse and a dog, so when it is tested with a new image it will hopefully identify the animal.

Machine learning

Some AI systems do not learn, for example, in a computer game the characters can move and interact with your character and adapt to what you do, but they are not learning what you do, they are following their pre-programmed rules.

Machine learning can be supervised, or unsupervised.

Supervised means that user is telling the program what its data means, for example, in the image recognition example the user says this image shows a horse.

KEY WORD

machine learning: a computer program that can adapt its stored rules or processes.

Unsupervised means that data is input, and then the program learns from the data. An example of this is clustering. The data is plotted on a graph, and then the program identifies which items of data are close to each other. In image recognition, the shapes in the images could be input and then the features plotted, the program can then group the images according to these features. This is done without any human interaction needed.

COMPUTER SCIENCE IN CONTEXT

Components in a computer game can make use of machine learning. For example, an AI chess player. The program will alter the moves that it makes based on the games it plays. This simulates how a human learns to play the game, they learn which moves work and which don't. This could be supervised learning if they are told which moves are the best to make. It could be unsupervised learning where it can play other people or computer programs, and change its own data and rules without being told what to do.

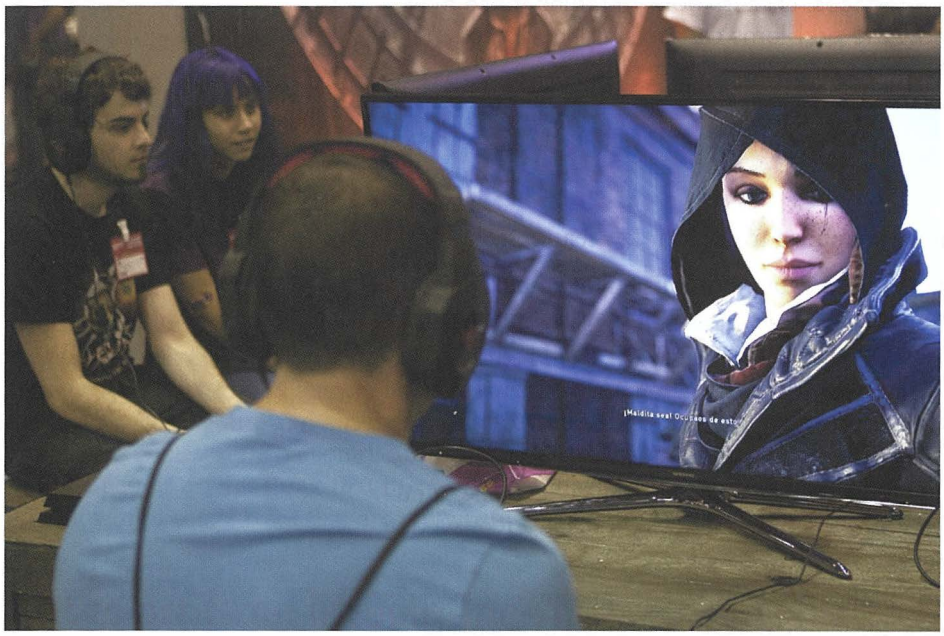


Figure 6.11: A person playing a computer game

ACTIVITY 6.3

Find and play a computer game where you are competing against the computer. Discuss how you think the computer's character was controlled. Did it just repeat the same movements over and over, or did it change what it did based on your move? Discuss how you think it was created and programmed.

REFLECTION

After completing Activity 6.3, it can be helpful to think about how you approached it and ways you could improve your research skills for future activities.

1 How did you approach this task? Did you play the computer game without thought to the character? Did you try lots of different moves to see how the computer reacted?

2 What programming knowledge did you use to discuss how the computer's character was programmed?

Expert systems

An **expert system** is a special type of system that attempts to emulate the expertise of a human, for example, an engineer, or doctor. The system asks the user questions to determine the solution or answer. Depending on each answer, the system will ask a different question.

For example, a program to help you fix a printer might first ask:

'Is there a green light on the printer?'

If the user says No, then they may be told, 'Press the button to turn the printer on.'

If the user says Yes, then they may be asked, 'Is there a red light underneath the paper picture?'

An expert system has four key features:

1 A **knowledge base**. This is a list of facts, for example, there is a green light below the power symbol.
2 A **rule base** that links the facts. For example, If the green light is on then the printer has power.
3 **Inference engine**. This is the part of the program that decides which question to ask next, or which answers it gives.
4 The user interface. This is the part that outputs questions and statements to the user, and allows the user to enter data.

KEY WORDS

expert system: a system that attempts to replicate the knowledge of an expert.

knowledge base: part of an expert system that stores the facts.

rule base: part of an expert system that stores the rules based upon the knowledge.

inference engine: part of an expert system that makes the decisions.

COMPUTER SCIENCE IN CONTEXT

Organisations often use expert systems to allow people to find help without having to communicate with people. These are usually repetitive problems that people may encounter. If you have a home internet connection and it is not working, then troubleshooting can tell you what actions to perform to fix it.

ACTIVITY 6.4

Create an expert system to determine the difference between a cat and a dog. Ask the user questions depending on their previous answers. This can be written as a computer program using a programming language of your choice, or as a flowchart showing which questions to move to depending on each answer.

Peer Assessment

Test each other's expert systems. How useable was the system? Were there any questions that you couldn't answer? Did it give you the correct answer?

Questions

10 Define the term artificial intelligence.

11 Describe what is meant by machine learning.

12 State the purpose of an expert system.

13 Give **two** examples of where expert systems can be used.

14 An expert system has an inference engine and interface. Describe the **two** other component parts of an expert system.

SUMMARY

An automated system performs actions without human intervention.
An automated system has sensors, a microprocessor and actuators.
Robotics is the creation and management of machines that perform actions that humans perform.
A robot contains a mechanical structure, electrical components including parts of an automated system, and it can be programmed to perform specific actions.
Robots can be found in a range of areas, for example, medicine and transport, and have advantages and disadvantages in each context.
Artificial intelligence is an area of computer science that aims to develop systems that mimic human intelligence.
Machine learning is one area of artificial intelligence where a program can amend its own data and algorithms.
An expert system attempts to simulate a human expert in a specific area.
An expert system is made up of a knowledge base, rule base, inference engine and interface.